

Wandi Xu

 Wandi Xu |  Wandi xu |  wandixu.com |  wxu68@jh.edu |  +1 (651)-395-8299

RESEARCH EXPERIENCE

Research Assistant

Oct 2020 – Jan 2022

**Institute on the Biology of Aging and Metabolism
University of Minnesota, Twin Cities**

Minneapolis, USA

Advisor: Prof. Paul Robbins and Prof. Lei Justan Zhang

Project One: Non-homologous end-joining (NHEJ) Assay

- Implemented cell passage, cell counting, cell transfection, and drug treatment

Project Two: The PCR Experiment

- Extracted RNA from the brain, lung, muscle, kidney and liver of mice, and converted it into cDNA
- Added primers to detect reaction data
- Analyzed data to evaluate whether the drug could slow down aging

Project Three: Self-mono-ADP-ribosylation Assay

- Optimized experimental conditions for drug concentration
- Observed the increasing activity of SIRT6 at different concentrations
- Conducted gel electrophoresis and analyzed gels using ImageJ

Research Assistant

Aug 2022 – present

Johns Hopkins University

Baltimore, MD, USA

Advisor: Prof. Hernandez Rigoberto

Project One: Structural and Dynamical Consistency in Lipid Vesicles across All-Atom, MARTINI, and Dissipative Particle Dynamics

- Built a 12 nm all-atom vesicle and multiscale coarse-grained vesicles (12–60 nm) using MARTINI and DPD; evaluated model transferability across scales.
- Validated structural and dynamical consistency through bilayer thickness, tail end-to-end distances, density/radial distributions, Van Hove functions, and scattering analyses.

Project Two: Binding Interactions between MPA-Coated Gold Nanoparticles and Proteins

- Performed all-atom MD simulations to characterize protein–nanoparticle interactions for multiple proteins (cyt c, aprotinin, ubiquitin, etc.).
- Identified binding orientations and key residues, and analyzed structural responses (RMSD, Rg, secondary structure) over 100–200 ns trajectories.
- Constructing all-atom models of MUA- and MUTAB-functionalized gold nanoparticles using CHARMM force field parameters; performing energy minimization, equilibration, and production MD simulations with GROMACS.
- Investigating the structural and dynamical properties of mixed-ligand nanoparticle surfaces as a function of ligand composition and solution conditions.

WORKING EXPERIENCE

New Oriental Vision Overseas Consulting Co. Ltd.

Position: Teaching Assistant

Beijing, China

Aug 2019 – Sep 2019

- Taught high school students how to prepare for the Scholastic Aptitude Test

Private School Education

Position: Instructor

Qinhuangdao, China

Oct 2017 – Aug 2018

- Tutored junior high students in mathematics, physics, and chemistry

Johns Hopkins

Position: Teaching Assistant

MD, USA

Aug 2022 – present

- Tutored for Intro Chem, General Chem, and Physical Chem courses

PUBLICATIONS

Deng, Chaoyi et al. (Aug. 2024). “Nanoscale Iron (Fe₃O₄) Surface Charge Controls Fusarium Suppression and Nutrient Accumulation in Tomato (*Solanum lycopersicum* L.)” In: *ACS Sustainable Chemistry & Engineering* 12.35, pp. 13285–13296. DOI: [10.1021/acssuschemeng.4c04800](https://doi.org/10.1021/acssuschemeng.4c04800).

Zhang, Lei et al. (Oct. 2024a). “Development of novel flavonoid senolytics through phenotypic drug screening and drug design”. In: *bioRxiv*. DOI: [10.1101/2024.10.27.620529](https://doi.org/10.1101/2024.10.27.620529).

– (Oct. 2024b). “Identification of lipid senolytics targeting senescent cells through ferroptosis induction”. In: *bioRxiv*. DOI: [10.1101/2024.10.14.618023](https://doi.org/10.1101/2024.10.14.618023).

Zhang, Lei et al. (June 2025). “Fucoidans are senotherapeutics that enhance SIRT6-dependent DNA repair”. In: *Research Square*. DOI: [10.21203/rs.3.rs-6613032/v1](https://doi.org/10.21203/rs.3.rs-6613032/v1).

Xu, Wandu et al. (2026). “Structural and Dynamical Consistency in Vesicles across All-atom, MARTINI and Dissipative Particle Dynamics”. In: *ACS Omega*. in revision.

Zhang, Lei Justan et al. (2026). “Polyunsaturated lipid senolytics exploit a ferroptotic vulnerability in senescent cells”. In: *Cell Press Blue* 1.1. DOI: [10.1016/j.cpblue.2026.100004](https://doi.org/10.1016/j.cpblue.2026.100004).

ACADEMIC SERVICE

Reviewed for *Scientific Reports*: 3 manuscripts.

SKILLS

Computer	Matlab, ChemDraw, Mestrenova, Image J, iBright, Microsoft Office Suite, Git, Python, GROMACS, LAMMPS, NAMD, VMD, Quantum Espresso, VESTA, Blender, Latex
Laboratory	Thermo Scientific NanoDrop, Cell Culture Hood, Microscope, qPCR
Hobby	Reading, Pencil Sketch, Watercolor, Hiking, Tennis, Badminton, Swimming, Snowboarding, Play video games, Acrylic painting